

Category	Hypothesis Test	Reference Distribution	Parameter	Test-Statistic	Parametric vs Nonparametric	Sub category	Nature of Hypothesis	Assumptions / Conditions
Numerical vs Categorical	Z-Test	Standard Normal Distribution (of Mean)	Mean	Z-Score	Parametric	One Sample Z-test	- H0: Population mean is same as sample mean - Ha: Population mean is not same as (or greater/lesser than) sample mean	1. Sample size is >= 30 2. Population SD is mandatory. 3. Sampling distribution is Normal. 4. Random sampling. 5. Independent datapoints. 6. Two samples can be of different size but must be i.i.d.
						Two Sample Z-test	- H0: Two samples have same means - Ha: Two samples have different (or greater/lesser) means	
	Z-Proportion	Standard Normal Distribution (of Proportion)	Proportion	Z-Statistic	Parametric	One Sample Z-Proportion test	- H0: Population proprtion is same as sample mean - Ha: Population proprtion is not same as (or greater/lesser than) sample mean	
						Two Sample Z-Proportion test	- H0: Two samples have same proportions - Ha: Two samples have different (or greater/lesser) proportions	
	T-test	Student's T Distribution	Mean	T-Statistic	Parametric	One sample T-test	- H0: Population mean is same as sample mean - Ha: Population mean is not same as (or greater/lesser than) sample mean	1. Sample size is < 30 2. Population SD is optional. 3. Homogeneity of variances.
						Two-Sample Independent T-test	- H0: Two samples have same means - Ha: Two samples have different (or greater/lesser) means	
						Two-Sample Paired T-test	- H0: No change in sample mean before and after the treatement - Ha: There is significant change in sample mean before and after treatement	
	ANOVA	F distribution	Mean	F-Ratio	Parametric	One-way ANOVA	- H0: No significant difference in mean among the groups - Ha: At least one of the group have significantly different mean	
						Two-way ANOVA	- H0: - Ha:	
	Kruskal-Wallis Test	Chi-square distribution	Median	H-Statistic	Nonparametric	NO	- H0: No significant difference in median among the groups - Ha: At least one of the group have significantly different median	NO
Categorical vs Categorical	Shapiro-Wilk Test	W distribution	NA	W-Statistic	Nonparametric	NO	- H0: Distribution of sample is Gaussian - Ha: Distribution of sample is NOT Gaussian	NO
	Levene's Test	F distribution	Variance	W-Ratio	Parametric	NO	- H0: All samples have similar Variance. - Ha: Al least one of the sample have significantly different variance.	NO
	Chi-Squared Test	Chi-square distribution	NA	Chi-Square Statistic	Nonparametric	Goodness of Fit test	- H0: No significant difference between the observed and expected results - Ha: Observed results are significantly different from expected results	
						Test for Independence	- H0: Two categorical features are independent - Ha: Two categorical features are NOT independent	
Numerical vs Numerical	Kolmogorov-Smirnov Test	NA	NA	D-Statistic	Nonparametric	NO	- H0: Both the distributions are same - Ha: Both the distributions are NOT same	NO
	Correlation Analysis (not an Hypothesis Test)	NA				Pearson correlation coefficient Spearman's rank correlation coefficient	NA	